

NAME: P.KAVYA SAI

ROLL NUMBER: 2420030322

LinkedIN submission: [LinkedIN Post](#)

Task 01 Documentation: Fundamentals of Prompt Engineering

Program: SkillCraft Technology

Objective: Learn the basics of prompt engineering by experimenting with clarity, specificity, and tone. Select three vague or ineffective prompts, rewrite them into highly structured versions using frameworks (Persona, Context, Task, and Constraints), and analyze the differences in Large Language Model (LLM) behavior.

Core Lesson: Vague inputs yield vague outputs. Shifting from open-ended questions to highly detailed instructions completely transforms the quality and utility of LLM responses.

Theoretical Framework

To engineering better prompts, we transform open-ended text into explicit instructions by mapping out four key pillars:

1. **Persona:** Defining who the AI is acting as (e.g., Expert Copywriter, Professional Employee).
2. **Context:** Providing background information, audience details, or current situations.
3. **Task:** Giving a clear, direct action verb defining what needs to be produced.
4. **Constraints:** Setting rules for format, character limits, tone, style, and what to avoid.

Prompt Experiments & Comparisons

Case Study 1: Social Media Content Creation

- **Vague Prompt:** "Write a tweet about AI" (See image_c22a5a.png)
- **Improved Prompt:**

"You're an expert social media copywriter who writes for thought leaders in education. You write highly engaging and compelling content. I am an AI educator & content creator with 100,000 subscribers. My audience is young Indian AI enthusiasts who don't want to depend on traditional education systems... Write an engaging tweet guiding students step by step about which AI skills to get a high-paying job right after college in India. Write in the crisp & thoughtful style of Tim Ferriss. Do not use emojis & exclamation. Keep sentences short & simple... Do not exceed 280 characters." (See image_c22a75.png)

Output Comparison & Analysis

Metric	Vague Prompt Output	Improved Prompt Output
Tone	Generic, inspirational, cliché ("collaborative partner").	Authoritative, cynical of traditional education, actionable, direct.
Specificity	High-level philosophy about human-AI collaboration.	Highly practical. Names concrete skills: API Integration, Vector Databases (Pinecone/Milvus), and Fine-tuning.
Formatting	Standard paragraphs with common hashtags.	Structured numbered list with clean formatting and a clear closing hook line: <i>"Proof of work beats a degree."</i>

Case Study 2: Workplace Communication

- **Vague Prompt:** "Write an email to my boss about taking time off."
- **Improved Prompt:**

"Act as a professional but approachable employee. Write a short email to my manager, Sarah, requesting 3 days of paid time off (PTO) from July 14th to July 16th for a family wedding. Mention that I have already asked Mike to cover my urgent client emails while I'm away, and that I'll finish up the monthly report before I leave. Keep it under 150 words." (See image_c22ab8.jpg)

Output Comparison & Analysis

Metric	Vague Prompt Output	Improved Prompt Output
Utility	Low. Provides a generic template littered with placeholders like [Boss's Name] and [start date].	High. Fully written and ready to send, customized with specific dates, names, and reasons.
Length & Flow	Long and broad to fit multiple generic use cases.	Exceptionally concise, tailored tightly under the requested 150-word constraint.

Metric	Vague Prompt Output	Improved Prompt Output
Contextual Edge	Proactively asks the user for more follow-up information.	Solves the problem outright by seamlessly integrating the coverage plan (Mike) and pending deliverables (monthly report).

Case Study 3: Academic Study & Research

- **Vague Prompt:** "Tell me about the Roman Empire."
- **Improved Prompt:**

"I am studying for a college-level introductory history exam. Create a structured study guide about the fall of the Western Roman Empire. Focus strictly on the primary political and economic causes between 300 AD and 476 AD. Use clear headings, bullet points for key facts, and include a 'Key Terms' section at the end with definitions for 5 crucial vocabulary words." (See image_c22d7c.jpg)

Output Comparison & Analysis

Metric	Vague Prompt Output	Improved Prompt Output
Scope	Massively overbroad. Covers origins, growth, religion, language, architecture, achievements, and legacy.	Deep and focused. Lasers in on the specific timeline (300\text{ AD} - 476\text{ AD}) and historical themes requested.
Structure	General overview sections that act as an encyclopedia entry.	Structured strictly as a targeted study guide split by explicit causes (e.g., "Weak and Unstable Leadership", "Administrative Division").
Pedagogical Value	Good for casual reading, poor for targeted active recall or exam prep.	Excellent. Separates "Key Facts" from "Why It Mattered" and cleanly compartmentalizes concepts for students.

☒ **Key Insights & Conclusion**

Through these three structured experiments, the foundational laws of Prompt Engineering became abundantly clear:

- **Constraints foster creativity and accuracy:** Restricting word limits, forbidden styles (e.g., eliminating emojis), or establishing fixed time horizons forces the LLM to drop filler content and deliver high-signal information.
- **Context drives personalization:** Providing the target audience parameters completely alters the vocabulary choice, examples generated, and tone used by the AI model.
- **The "Persona" hack:** Giving the LLM a professional background allows it to simulate real-world contextual nuances, such as handling operational handoffs in business emails or picking modern technical tools for a roadmap.